

Literature Review & Policy Context: Electric Vehicle (EV) Adoption and Charging Infrastructure Planning in Washington State

1. Policy Framework and State Goals

Washington State has set ambitious targets to accelerate electric vehicle (EV) adoption as part of its Clean Energy Transformation Act and 2035 Clean Transportation Vision.

The state aims to reach:

- 45% EV adoption by 2030
- 70% EV adoption by 2040
- 95% EV adoption by 2050
- 80% residential home/residential-charger adoption by 2030 – most cost saving way for users.

These milestones are central to Washington's plan for achieving a carbon-neutral transportation sector. They highlight two challenges: forecasting future EV adoption and optimizing charging infrastructure across urban, suburban, and rural counties.

The goal of this capstone project is to model and forecast EV adoption trajectories at the county level (2025–2050) and estimate corresponding public-charging infrastructure needs aligned with these policy goals.

Resource: https://governor.wa.gov/sites/default/files/exe_order/21-04%20-%20Zero%20Emission%20Vehicles.pdf

<https://evchargen.com/articles/2030-projection-home-charging-to-fulfill-80-of-electric-vehicle-charging-needs>

2. Review of Prior Research and Modeling Approaches

Although Tesla does not publish formal research on its charging-infrastructure algorithms, a growing body of academic and governmental work offers comparable frameworks for EV adoption forecasting and charger deployment optimization.

2.1 Forecasting EV Adoption

- Kamis et al. (2024) — Predictive Models of Electric Vehicle Adoption in the United States

Developed machine-learning and regression models to predict adoption using socioeconomic features (income, density, incentives).

 <https://www.sciencedirect.com/science/article/pii/S2590198224000277>

- Ramadoss et al. (2025) — Classifying Electric Vehicle Adopters and Forecasting Adoption
Applied a Bass diffusion model segmented by adopter types. The model fits non-linear growth curves and can be constrained to match state policy targets (e.g., 45%, 70%, 95%).
<https://www.nature.com/articles/s44333-025-00049-1>
- Archsmith, Muehlegger, & Rapson (2021) — Future Paths of Electric Vehicle Adoption in the United States
Explored long-term adoption trajectories influenced by pricing and incentives. Their regression-diffusion hybrid can inform Washington’s county-level projections.
https://rapson.ucdavis.edu/uploads/8/4/7/1/84716372/ev_forecasting_main.pdf
- Cervini et al. (2024) — Temperature and Electric Vehicle Adoption: A ZIP Code Level Study
Identified climate as a significant factor affecting BEV vs. PHEV adoption. This could guide adjustments for Washington’s diverse climates (King vs. Chelan County).
<https://www.sciencedirect.com/science/article/abs/pii/S1361920924003924>

2.2 Charging Infrastructure Planning

- Kaushik et al. (2025) — Electric Vehicle Charger Infrastructure Planning
Introduced a two-phase framework combining spatial demand estimation with grid-aware coverage optimization.
<https://arxiv.org/abs/2509.23699>
- Wang et al. (2025) — Spatiotemporal Planning of Electric Vehicle Charging Infrastructure
Developed a three-stage model: forecasting demand, planning site locations, and phasing rollout.
<https://pmc.ncbi.nlm.nih.gov/articles/PMC12496191/>
- Junker et al. (2025) — Data-Driven Optimization of EV Charging Station Placement Using Causal Discovery
Utilized real-world charger utilization data to infer causal relationships and optimize station placement.
<https://arxiv.org/abs/2503.17055>
- Li et al. (2025) — DOVA-PATBM: An Intelligent, Adaptive, and Scalable Framework for Optimizing Large-Scale EV Charging Infrastructure
Proposed a hierarchical, graph-based optimization model integrating geographic clusters and traffic data.
<https://arxiv.org/abs/2506.15289>
- Urban Public Charging Infrastructure Planning for EVs (2025)
Introduced continuum approximation models, representing charger density as continuous spatial variables for dense urban areas.
<https://www.sciencedirect.com/science/article/pii/S136192092500015X>

3. Gaps in Existing Literature

- Policy Integration: Few studies explicitly calibrate models to state-mandated adoption targets (45%, 70%, 95%).
- Residential Charging Interaction: Most literature isolates public charging; few incorporate home-charger penetration as a dynamic offset.
- Regional Resolution: National studies model at state or metro level, not county-level spatial resolution.
- Long-Term Forecast Horizon: Few studies extend forecasts beyond 2035.

4. Proposed Approach (Policy-Aligned Modeling Path)

1. Forecast EV Adoption (2025–2050): Use logistic or Bass-diffusion models constrained to Washington’s milestones; include demographic, income, and charger-density predictors.
2. Estimate Public-Charging Needs: Apply benchmark ratios (1 per 1,500 urban, 1 per 2,500 suburban, 1 per 5,000 rural), adjusting for 80% residential charging by 2030.
3. Spatial Allocation: Optimize charger placement using facility-location algorithms; integrate interstate and population clusters.
4. Scenario Simulation: Compare baseline, policy-aligned, and home-charger heavy cases, and evaluate against state adoption thresholds.

5. References

- Kamis, A. et al. (2024). Predictive Models of Electric Vehicle Adoption in the United States.
<https://www.sciencedirect.com/science/article/pii/S2590198224000277>
- Ramadoss, P. et al. (2025). Classifying Electric Vehicle Adopters and Forecasting Adoption.
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- Kaushik, H. D. et al. (2025). Electric Vehicle Charger Infrastructure Planning.
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